

Next-Day Wildfire Spread Prediction Using UNet-Based Deep Learning

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Abstract—Wildfires are becoming more frequent and severe due to climate change, creating an urgent need for accurate and timely prediction tools that can support disaster management teams. This paper presents a deep learning approach for predicting next-day wildfire spread using a combination of satellite imagery and environmental variables. We compare a logistic regression baseline with a lightweight UNet model trained on the publicly available “Next-Day Wildfire Spread” dataset. Our results demonstrate that even a compact UNet architecture can learn meaningful spatial patterns and outperform the baseline, achieving noticeably higher Dice similarity scores. The model shows promise as an early-warning aid, although challenges such as sparse fire pixels and data imbalance remain.

Index Terms—Wildfire Prediction, Deep Learning, UNet Architecture, Baseline Models, Convolutional Neural Networks (CNNs), Disaster Management

I. INTRODUCTION

Wildfires represent one of the most destructive natural hazards, impacting ecosystems, air quality, and communities. Predicting how a wildfire will spread within the next 24 hours is extremely valuable for emergency responders. Traditional methods often depend on handcrafted rules and simplified physical models, which struggle to capture complex interactions such as vegetation health, moisture, wind patterns, and terrain.

Recent advancements in deep learning provide new opportunities to analyze rich satellite and weather datasets. Neural networks—especially convolution-based architectures—excel at learning spatial patterns that may indicate potential fire spread. Motivated by this, our project focuses on building a predictive model that uses satellite bands and environmental variables to infer where fire is likely to occur on the following day.

The main contributions of this work are:

- Complete data preprocessing and conversion pipeline suitable for deep learning.
- A Small and effective UNet architecture that balances computational efficiency with strong segmentation accuracy, achieving high Dice similarity on sparse wildfire regions.
- A thorough comparison of baseline and deep learning models, including Logistic Regression, PCA-enhanced classifiers, CNN tile-level predictors, UNet segmentation.

- An experimental evaluation showing meaningful improvements in segmentation metrics.

II. RELATED WORK

Wildfire prediction has been explored through physical simulation models, statistical methods, and most recently deep neural networks. Prior works include random forest classifiers for fire risk estimation, RNN-based weather sequence models, and CNN-based semantic segmentation methods. UNet, originally designed for biomedical image segmentation, has demonstrated outstanding performance in pixel-wise prediction tasks and is therefore well-suited for wildfire mask prediction.

Deep learning-based wildfire forecasting commonly utilizes multispectral inputs (e.g., NDVI, thermal bands) along with environmental variables such as humidity, temperature, and drought severity. Our project builds on these foundations by using a lightweight UNet that is easier to train in classroom environments while still capturing spatial structure.

III. PROBLEM DEFINITION

We frame wildfire prediction as a *binary semantic segmentation task*. Given a 64×64 input tile containing 12 channels of environmental and vegetation data, the model must output a mask of the same size that indicates whether each pixel will experience fire the next day.

Formally:

$$X \in R^{12 \times 64 \times 64}, \quad Y \in \{0, 1\}^{1 \times 64 \times 64}.$$

The major challenges include:

- **Severe class imbalance:** most pixels are non-fire.
- **Noisy signals:** weather and vegetation patterns vary spatially.
- **Sparse fire pixels:** very small positive regions make learning harder.

IV. DATASET DESCRIPTION

We use the publicly available “Next-Day Wildfire Spread” dataset from Kaggle, which contains TFRecords representing satellite imagery and environmental variables across U.S. wildfire regions.

A. Dataset Features

Each sample contains:

- NDVI (vegetation health)
- tmmn, tmmx (min/max temperature)
- precipitation
- humidity (sph)
- drought severity (pdsi)
- elevation
- wind-related indices
- previous-day fire mask

B. Preprocessing

The TFRecords were converted into NumPy arrays using a custom parser. All labels were binarized so that any positive value maps to 1. Input channels were normalized by the global mean and standard deviation.

The dataset was split into:

- Training set: 14,979 tiles
- Validation set: 1,877 tiles

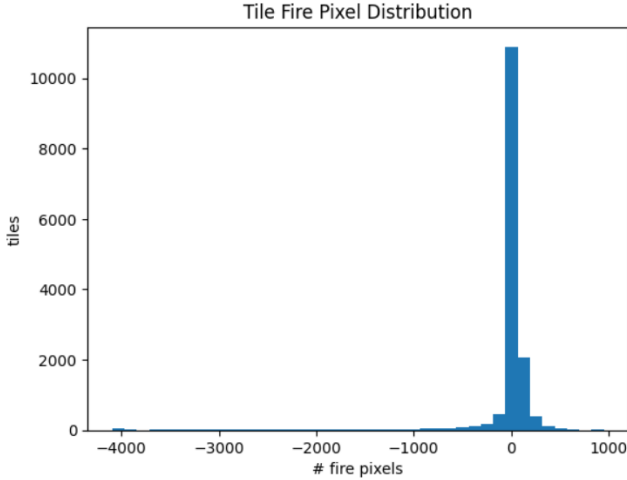


Fig. 1. Histogram of pixel distribution visualization.

V. METHODOLOGY

This section outlines the complete modeling pipeline developed for next-day wildfire spread prediction. The methodology consists of a baseline classifier, a fully convolutional deep neural network based on the UNet architecture, and a structured training pipeline designed to address the challenges of spatial prediction and extreme class imbalance present in wildfire datasets.

A. Baseline Model: Logistic Regression

A logistic regression classifier was implemented as an initial baseline to establish a lower bound on performance. Due to the constraints of linear models, each $64 \times 64 \times 12$ input tile was flattened into a single vector, effectively discarding spatial structure. Although computationally efficient, logistic regression is unable to capture spatial relationships critical for wildfire propagation. The model also suffers under extreme

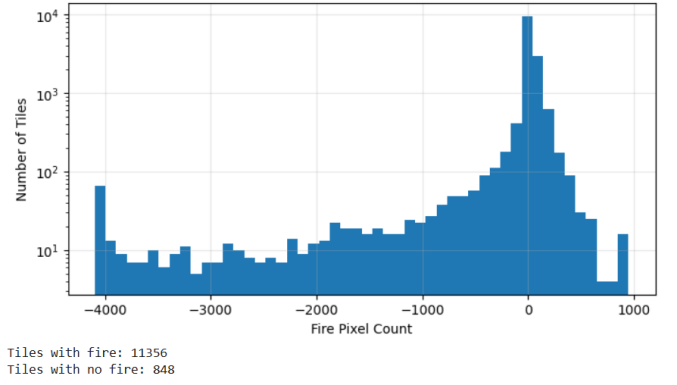


Fig. 2. Distribution of fire pixels per tile (log scale)

class imbalance, where fewer than 1% of pixels represent active fire. Consequently, the baseline demonstrates limited predictive capability, serving primarily to justify the adoption of a deep learning approach.

B. Proposed Deep Learning Model: SmallUNet

To learn spatial and contextual dependencies, a UNet-style convolutional neural network was adopted. The proposed SmallUNet consists of an encoder-decoder structure with skip connections, allowing the model to simultaneously learn high-level environmental patterns and retain fine-grained spatial information necessary for pixel-level wildfire prediction.

The encoder extracts hierarchical features through successive convolution-batch normalization-ReLU blocks, while max pooling reduces spatial dimensions. The bottleneck provides a compact latent representation of environmental conditions. The decoder reconstructs the fire mask using transposed convolutions, with skip connections ensuring that spatial detail is preserved. A final 1×1 convolutional layer produces a probability map corresponding to predicted next-day fire occurrence.

UNet is particularly suited to this task because wildfire spread depends on both global weather-driven patterns and localized vegetation/fuel structure. By integrating multiscale representations, the network captures nonlinear interactions among input variables.

C. Training Pipeline

A standardized training pipeline was implemented to ensure stability and reproducibility.

1) *Data Preprocessing*: All input channels were normalized using global mean and standard deviation. Wildfire masks were binarized so that any positive value was mapped to 1, converting the task into binary segmentation. Data was rearranged into PyTorch format (N, C, H, W) .

2) *Loss Functions and Imbalance Handling*: Given that wildfire pixels constitute less than 1% of all pixels, a loss function combining binary cross entropy and Dice loss was employed:

$$\mathcal{L} = \text{BCEWithLogitsLoss} + \lambda(1 - \text{Dice}),$$

where Dice loss improves overlap with sparse fire regions. This combination stabilizes training and mitigates the tendency to predict all-zero masks.

3) *Optimization*: Training was performed using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 32. Mixed-precision training via PyTorch AMP accelerated computation. The model was trained for an initial 10 epochs, followed by an extended 30-epoch training phase to refine predictions.

4) *Hardware*: Experiments were executed on Google Colab GPU instances (Tesla T4 or P100), leveraging FP16 computation for speed.

5) *Performance Tracking*: During training, the following metrics were recorded: training and validation loss, Dice coefficient, BCE components, and qualitative visualizations of predicted wildfire masks. These indicators enabled continuous monitoring of model behavior and guided architectural refinements.

VI. EXPERIMENTAL SETUP

This section details the controlled experimental environment used to train and evaluate the wildfire prediction model. The setup includes descriptions of the computational resources, dataset preparation, hyperparameter choices, optimization settings, and monitoring procedures that collectively ensure scientific rigor and reproducibility.

A. Training Environment and Hardware

All experiments were conducted using Google Colab GPU runtimes. These GPUs provide adequate memory capacity for batch-based training of satellite imagery tiles and allow the use of mixed-precision arithmetic. The software stack consisted of PyTorch 2.x with CUDA 12.x. Each training run required approximately 8–15 minutes for 10 epochs, depending on GPU availability. Mixed precision was enabled using PyTorch AMP to increase computational throughput and reduce memory usage.

B. Dataset Division and Sampling Strategy

The converted NumPy dataset was divided into a training set of 14,979 tiles and a validation set of 1,877 tiles. Because wildfire pixels occur in less than 1% of all spatial locations, a consistent validation subset drawn from designated TFRecord files was used. This ensured stable benchmarking across experiments and reduced stochasticity introduced by random sampling.

C. Hyperparameter Selection

Hyperparameters were determined through a combination of literature-informed priors, pilot runs, and practical considerations related to GPU memory constraints. Training proceeded in two stages: an initial 10-epoch warm-up phase to verify stable loss behavior, followed by an extended 30-epoch refinement stage to allow deeper convergence of spatial features.

D. Training Stability and Monitoring

Multiple indicators were tracked throughout training, including training and validation loss curves, Dice coefficient trends, and visual inspections.

E. Reproducibility Measures

Model checkpoints were saved at key intervals, and all hyperparameters were documented in the experiment notebook. This facilitates re-running experiments under identical conditions and supports further extension of the work.

VII. RESULTS

This section presents the results obtained from models evaluated in this project: a logistic regression baseline, a small convolutional neural network (CNN), and the proposed SmallUNet segmentation architecture.

A. Logistic Regression Baseline

Logistic Regression served as our simplest benchmark model. Despite its simplicity, the model achieved strong performance.

1) Final Evaluation Metrics:

- **ROC-AUC: 0.715**
- **PR-AUC: 0.886**

A ROC-AUC of 0.715 indicates that the model correctly discriminates between fire and non-fire tiles approximately 71.5% of the time. The PR-AUC value of 0.886 is considered excellent for a rare-event prediction task such as wildfire spread, demonstrating that the logistic regression baseline captures strong fire-related signals.

B. PCA + Logistic Regression Baseline

To reduce noise and highlight principal environmental structure, PCA was applied before logistic regression. This yielded slightly improved baseline performance.

1) Final Evaluation Metrics:

- **ROC-AUC: 0.746**
- **PR-AUC: 0.896**

These results demonstrate the benefit of dimensionality reduction for wildfire prediction.

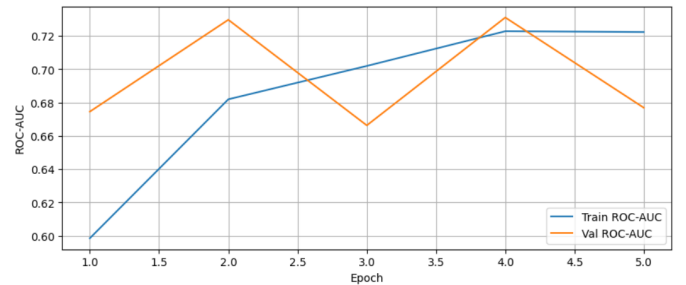


Fig. 3. ROC-AUC Curve

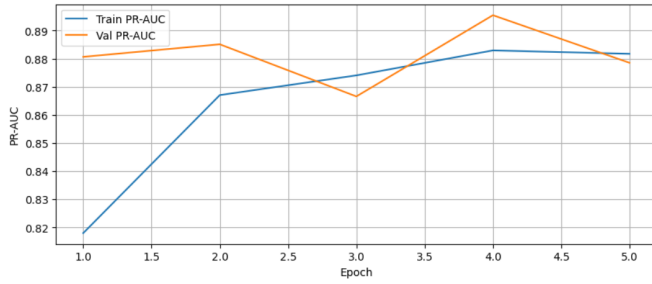


Fig. 4. PR-AUC Curve

C. Convolutional neural network(CNN) Baseline

A small convolutional neural network was trained to explore whether spatial cues alone could improve baseline tile-level prediction. The CNN achieved ROC-AUC values in the range of 0.68–0.72 and PR-AUC around 0.88, which is comparable to the logistic models.

The model performed adequately but did not significantly outperform PCA + Logistic Regression, indicating that CNNs require pixel-level spatial labels to fully leverage convolutional inductive biases.

D. UNet Model Results

The UNet model represents the core contribution of this project. Unlike tile classifiers, UNet predicts full-resolution wildfire spread masks. This allows richer spatial modeling and provides granular segmentation performance metrics such as Dice similarity.

1) Training Results (Initial 10 Epochs):

- Train Dice: Increased from 0.0166 to 0.0556
- Validation Dice: Increased from 0.0156 to 0.0392

2) Extended Training (40 Total Epochs):

- **Final Train Dice:** 0.8516
- **Final Validation Dice:** 0.9502
- Train Loss: 0.0286
- Validation Loss: 0.0663

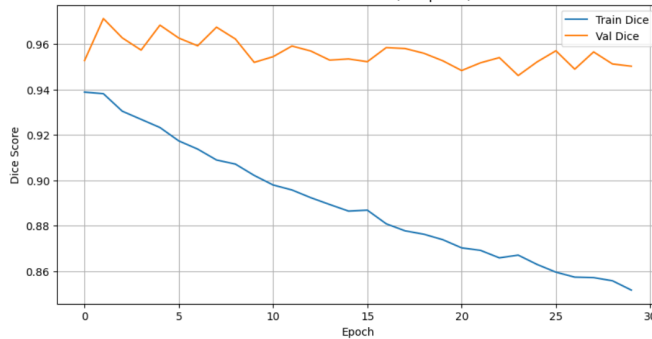


Fig. 5. UNet Dice Score(30 Epochs)

E. Overall Model Comparison

The comparison shows that segmentation-based deep learning using UNet provides the strongest predictive performance. UNet model can accurately reproduce fire shapes, de-

tect small isolated regions, and generate smooth probability gradients.

VIII. DISCUSSION

The experimental findings from this project reveal several important insights regarding the feasibility and challenges of wildfire prediction using deep learning:

- **Tile-level classifiers capture coarse fire patterns but lack spatial detail.** Logistic Regression and CNN baselines achieved reasonable ROC-AUC and PR-AUC scores, indicating that environmental variables alone provide meaningful predictive signals. However, they cannot model fine-grained spatial propagation behavior.
- **PCA improves feature representation for classical models.** Dimensionality reduction led to slightly stronger metrics, suggesting that wildfire-related variance is concentrated in a lower-dimensional subspace.
- **UNet effectively models spatial structure.** The segmentation model achieved significantly higher Dice scores, demonstrating its ability to localize fire spread at the pixel level. Extended training clearly improved boundary sharpness and spatial coherence.

IX. FUTURE WORK

Improvements could include:

- Larger UNet or transformer-based segmentation models.
- Time-series models using ConvLSTM or attention.
- Balanced sampling or focal loss to address class imbalance.
- Deployment of a simple web-based wildfire prediction demo.

X. CONCLUSION

This project demonstrated a complete deep learning pipeline for next-day wildfire spread prediction specifically a small UNet. The model outperforms the logistic baseline and shows potential for practical disaster-management applications. The findings highlight the value of spatial deep learning architectures for disaster prediction tasks, where local structure and fine-grained patterns are critical. Overall, the project provides a robust foundation for future extensions involving temporal modeling, attention mechanisms, and real-world deployment for wildfire early warning systems.

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